

Understanding Data Scarcity in Foundation-Model-Based Histopathology Patch Classification

Yannik Queisler

Thesis Proposal

October 2, 2026

1 Motivation

Pathology foundation models, pretrained with self-supervision on large slide collections, have become the default way to represent histopathology patches (Chen et al., 2024; Vorontsov et al., 2024). Public models are now compared systematically on clinical tasks (Campanella et al., 2025), yet such benchmarks vary the model and fix the downstream data. How much labelled data a classifier on top of these representations needs is much less understood, although labels are expensive (Lu et al., 2021) and thinnest for the rare cancers these models are promoted for (Vorontsov et al., 2024).

Nominal sample counts are a poor measure of data sufficiency. Patches are nested within patients, and mixing them across training and evaluation inflates reported scores by up to 41% (Bussola et al., 2021); thousands of patches may carry the information of a few dozen patients. Class frequencies are also strongly uneven, which in medical imaging distorts both discrimination and calibration (Mosquera et al., 2024), and long-tailed recognition work separates representation from classifier training (Kang et al., 2020). With frozen features the representation is fixed, so the imbalance effect can be studied on the classifier alone. Clinical use also needs well-calibrated probabilities, which modern networks often lack (Guo et al., 2017).

This thesis therefore separates three forms of scarcity: skewed class allocation, too few patches, and too few patients. It asks which of them drives the performance loss of classifiers trained on foundation-model representations, why, and how far algorithms can compensate, telling practitioners where annotation effort pays off and when a better algorithm cannot replace it.

2 Research Questions

1. **Class imbalance.** How does it affect discrimination and calibration, and which mechanisms cause the loss?
2. **Patch shortage.** How does it affect performance when patient coverage is held constant?
3. **Patient shortage.** How does it affect performance independently of patch count, and which properties of the learned feature distributions explain the loss?
4. **Mitigation.** Which forms of scarcity can be mitigated algorithmically without acquiring more data?

3 Methodology

Controlled experiments use TCGA-UT (Komura et al., 2022) (patches from 32 cancer types) and BRACS (Brancati et al., 2022) (breast lesions including atypical categories) with frozen

pathology foundation-model representations. Three factors are manipulated separately while the others are controlled (Figure 1):

- **Class allocation:** how the training data is distributed across classes.
- **Patch count:** patches per patient, with patient coverage held constant.
- **Patient count:** number of independent patients, with patch count held constant.

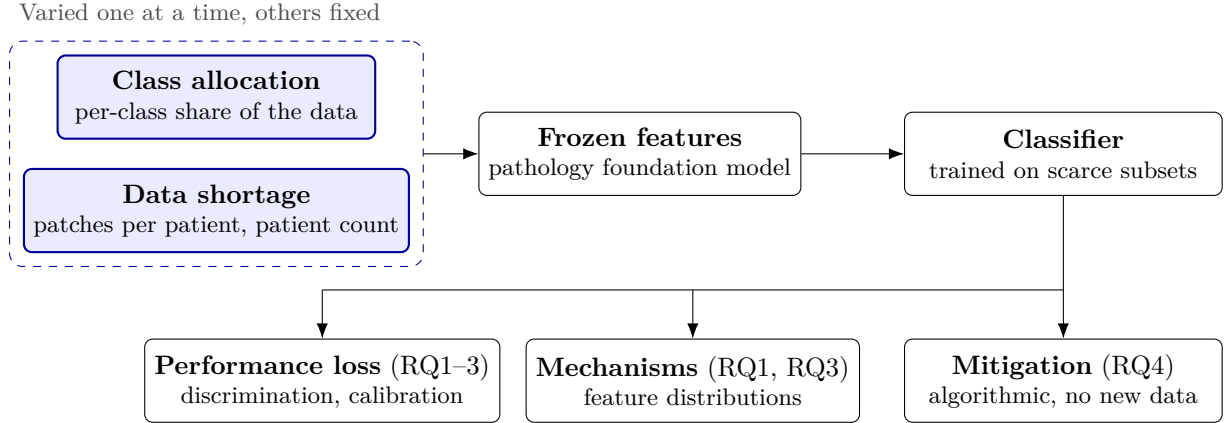


Figure 1: Experimental design. Each scarcity factor (blue) is varied alone while the others stay fixed.

Every setting is scored on discrimination and calibration. Mechanism analyses look at class influence, tail support and how well class-level patient distributions are estimated in feature space. Representative mitigation approaches are then tested against the loss each factor causes.

4 Preliminary Findings

First experiments show that class imbalance costs frozen-feature probes little on TCGA-UT at first and then accelerates (balanced-accuracy loss 0.6 pp at a largest-to-smallest class ratio of 10, 2.9 pp at 100), but starts immediately and is 2.6 times larger on BRACS (7.7 pp at 100). Unequal class influence and reduced tail support both contribute and compound at high ratios. Calibration error rises, but one global temperature removes 92–96% of the increase. Patient shortage produces a distinct loss, caused mainly by poor estimation of class-level patient distributions, and increasing patch depth cannot fully substitute for patient breadth. Mitigation of patient-shortage damage has so far been limited.

5 Expected Contribution

The thesis gives a systematic account of how **class allocation**, **patch support** and **patient support** contribute differently to data scarcity in computational pathology. The resulting framework should clarify three things:

- when additional nominal samples help,
- when independent patient diversity matters more,
- when algorithmic mitigation can compensate for limited data.

References

- Nadia Brancati, Anna Maria Anniciello, Pushpak Pati, Daniel Riccio, Giosue Scognamiglio, Guillaume Jaume, Giuseppe De Pietro, Maurizio Di Bonito, Antonio Foncubierta, Gerardo Botti, Maria Gabrani, Florinda Feroce, and Maria Frucci. BRACS: A dataset for BReAst carcinoma subtyping in H&E histology images. *Database*, 2022:baac093, 2022. doi: 10.1093/database/baac093.
- Nicole Bussola, Alessia Marcolini, Valerio Maggio, Giuseppe Jurman, and Cesare Furlanello. AI slipping on tiles: Data leakage in digital pathology. In *Pattern Recognition. ICPR International Workshops and Challenges*, 2021. URL <https://arxiv.org/abs/1909.06539>.
- Gabriele Campanella et al. A clinical benchmark of public self-supervised pathology foundation models. *Nature Communications*, 16, 2025. doi: 10.1038/s41467-025-58796-1.
- Richard J. Chen, Tong Ding, Ming Y. Lu, et al. Towards a general-purpose foundation model for computational pathology. *Nature Medicine*, 30(3):850–862, March 2024.
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, pages 1321–1330. PMLR, 2017.
- Bingyi Kang, Saining Xie, Marcus Rohrbach, Zhicheng Yan, Albert Gordo, Jiashi Feng, and Yannis Kalantidis. Decoupling representation and classifier for long-tailed recognition. In *International Conference on Learning Representations*, 2020. URL <https://openreview.net/forum?id=r1gRTCvFvB>.
- Daisuke Komura, Akihiro Kawabe, Kenta Fukuta, Kazuya Sano, Tsukasa Umezaki, Haruka Koda, Riku Suzuki, Keisuke Tominaga, Michiaki Ochi, Hirofumi Konishi, et al. Universal encoding of pan-cancer histology by deep texture representations. *Cell Reports*, 38(9):110424, 2022. doi: 10.1016/j.celrep.2022.110424.
- Ming Y. Lu et al. Data-efficient and weakly supervised computational pathology on whole-slide images. *Nature Biomedical Engineering*, 5(6):555–570, 2021. doi: 10.1038/s41551-020-00682-w.
- Candelaria Mosquera, Luciana Ferrer, Diego H. Milone, Daniel Luna, and Enzo Ferrante. Class imbalance on medical image classification: towards better evaluation practices for discrimination and calibration performance. *European Radiology*, 34(12):7895–7903, 2024. doi: 10.1007/s00330-024-10834-0.
- Eugene Vorontsov, Alican Bozkurt, Adam Casson, George Shaikovski, Michal Zelechowski, Kristen Severson, Eric Zimmermann, et al. A foundation model for clinical-grade computational pathology and rare cancers detection. *Nature Medicine*, 30(10):2924–2935, 2024. doi: 10.1038/s41591-024-03141-0.